The Gold Nanorod Remote Walking Device

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Introduction

Although scaling human technology up in size is relatively straightforward, scaling down mechanisms to the molecular scale without losing functionality remains a major challenge.

Movement at Molecular Scale

- "Swimming" is inefficient at molecular scale because of the dominance of stochastic forces over inertia.
- Devices that “walk,” using limbs to grip, pull, and push for locomotive force, are of interest at this scale because of their predictable motion and energy efficiency.

Biological Inspiration

- Billions of years of evolution have provided inspiration on molecular-scale walking devices.
- Myosin [2] and Kinesin [3] are “walking” objects found in the human body that perform multiple functions, including carrying payloads thousands of times their size.

Current Progress

- Research into “nanowalkers” has made progress, albeit with many limitations. Looking widely, important characteristics of these walkers include autonomy, fuel, and track design.
- Previously designed nanowalkers have demonstrated remote power and control but have required sophisticated, scale-constrained tracks [4, 5].
- We propose a remotely powered and controlled walker that possesses the ability to “walk” on a much simpler track that attracts the walker's feet with an electric field.

References

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[4] Sherman, W. B.; Seeman, N. C. A precisely controlled DNA biped walking device. 2004, 4, 1203–1207.

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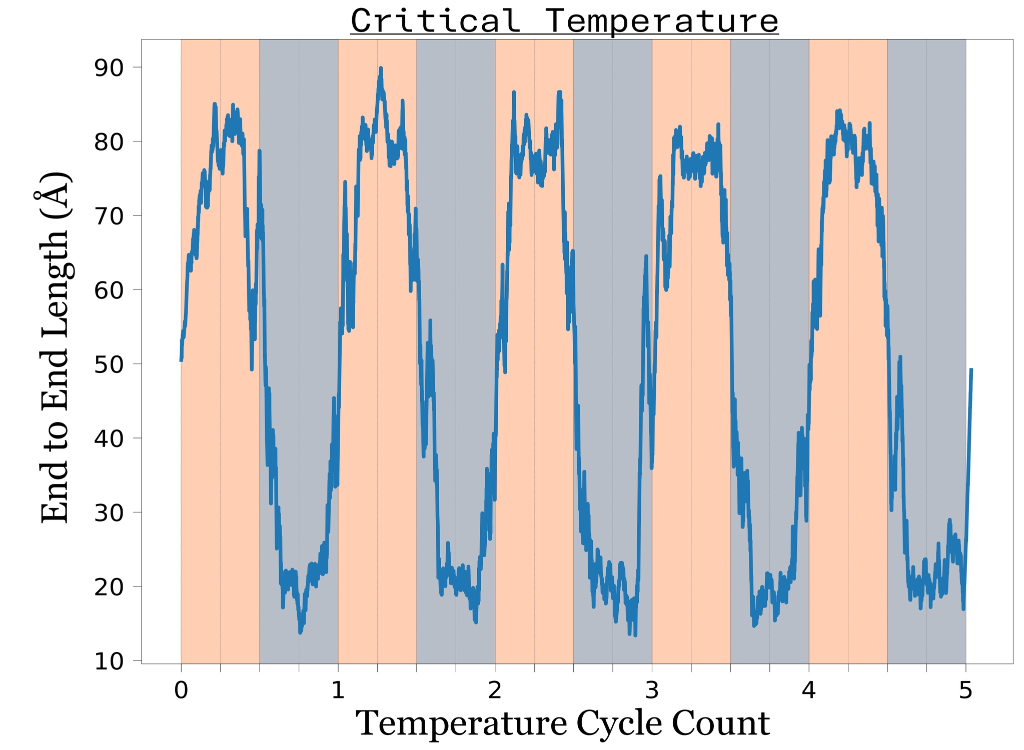
Methods

The Nanorod Walking Device takes advantage of two well-studied mechanisms at the molecular scale.

Temperature Response of Elastin-Like-Proteins

- Elastin-Like-Proteins are a class of proteins defined by their strong temperature response.
- Above their critical temperature, the proteins exhibit a collapse that reduce their end-to-end distance by up to 80%.
- We can assemble these proteins into a brush to create a "leg" whose extension is controlled by our machine's temperature.

Effective Length of VPGVG Brush Above and Below Critical Temperature



We simulate temperature induced transitions between temperatures colder (blue) and hotter (orange) than our protein's critical temperature. Our simulation successfully reproduces temperature response for our chosen elastin-like-protein, VPGVG.

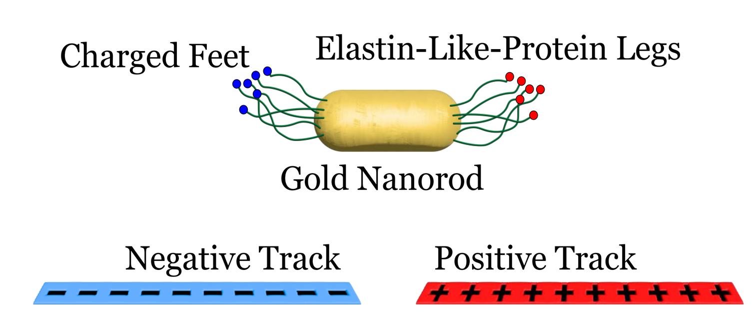
Selective Absorption by Gold Nanorods

- Selective absorption of radiation by nanoparticles in liquids has been demonstrated for gold clusters less than 100 nm in diameter.
- Studies have shown that a laser pulse can increase local temperature of nanoparticles by 800 K in less than 100 ps [1].
- Thus, the laser pulsing of nanoparticles is a convenient method for control the temperature of a small nano-scale machine.

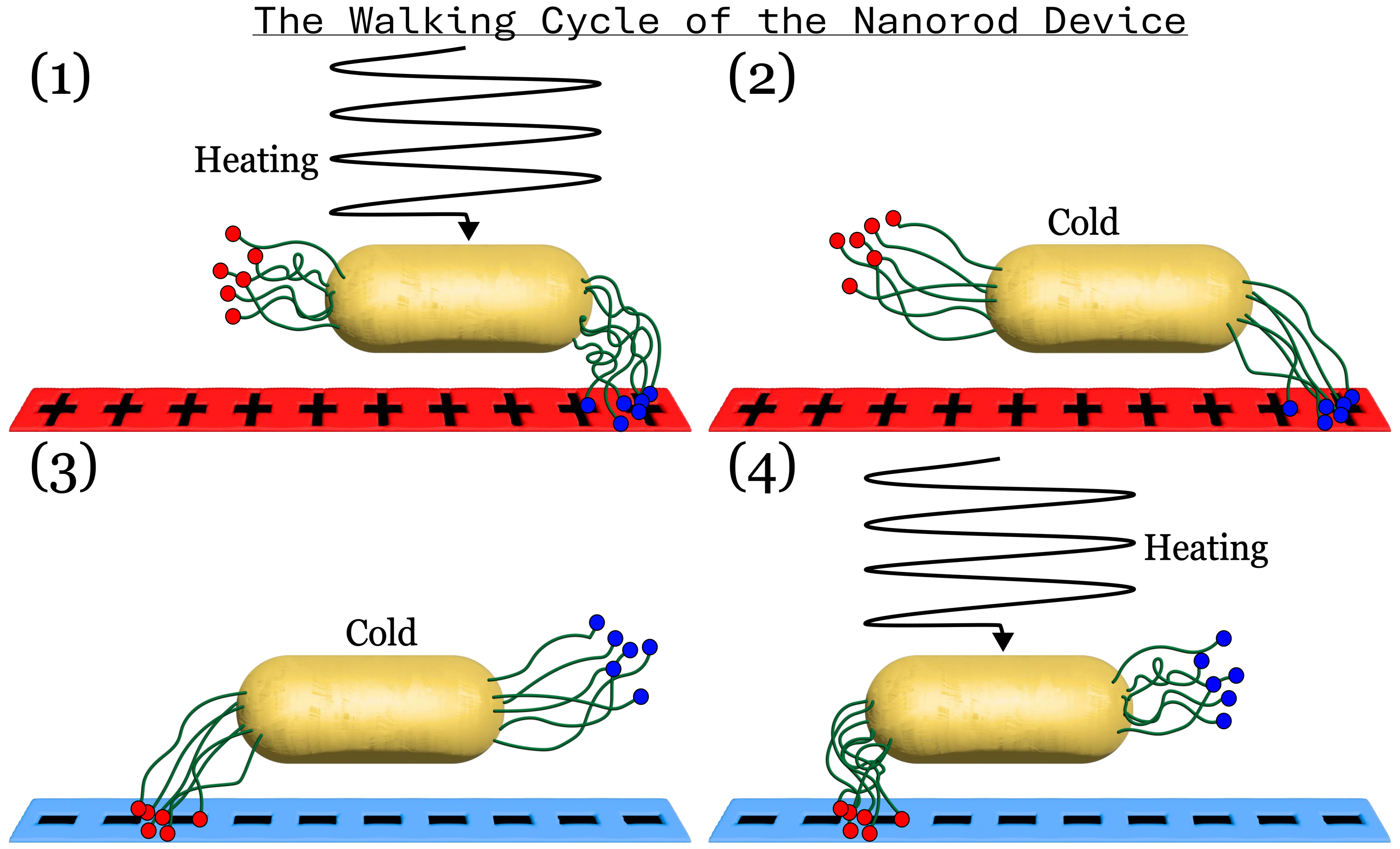
Simulation of The Device

- Potentials that characterize behavior between amino acids were used to simulate the temperature response of our "legs."
- Charged amino acids (feet) were attached to the legs, and the legs were then attached to a realistically sized gold nanorod.
- Temperature was controlled manually based on the nanorod's expected selective absorption response.
- Our track is simulated with an oscillating charge to attract the oppositely charged feet at different points of the operational cycle.

(Digitally) Assembled Device



The Walking Cycle of the Nanorod Device

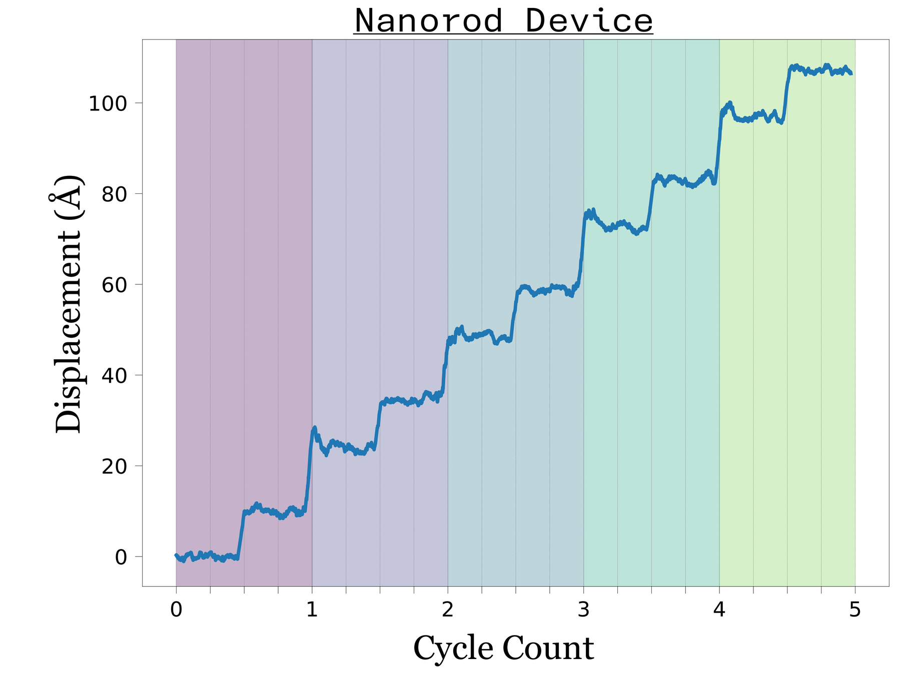


Walking Cycle Diagram:
(1) The track's field is negative and heating is applied to raise temperature above critical temperature of proteins. The legs collapse and the right (negative) legs are attracted to the track.
(2) Heating is paused and the legs extend as they cool below the critical temperature. This step produces a push to the left.
(3) The track electric field is reversed to attract the left (positive) legs.
(4) Heating is applied again, and the legs collapse as they exceed the critical temperature. This step produces a pull to the left.

Results

- Our design was simulated to reliably make forward progress.
- Simulations swept parameters such as leg length, nanorod size, track charge, and cycle times to optimize speed and reliability.
- The number of options for each parameter was heavily limited by access to computing time, but the most optimal combination found was chosen.

Displacement over Five Walking Cycles of the Nanorod Device



The simulation of the device successfully "stepped" like a staircase reliably over many steps. The "pull" step (4) was found to be slightly more effective than the "push" (2) step

Conclusions

- Our design is promising for remote-controlled, remotely-powered, nano-scale walking devices, although this technology is yet to be experimentally reproduced.
- Future work should focus on testing and improving the device's performance in less empty environments like biological fluid.

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